

Toward Grounded YOLO-SAM: Unified Detection–Segmentation Framework for Agricultural Intelligence

Pooja Verma
Advanced Technology
Development Centre
Indian Institute of Technology
Kharagpur
Kharagpur, India
pverma@kgpian.iitkgp.ac.in

Ayan Paul
Department of Agriculture and
Food Engineering
Indian Institute of Technology
Kharagpur
Kharagpur, India
ayanpaul2210@kgpian.iitkgp.ac.in

Rajendra Machavaram
Department of Agriculture and
Food Engineering
Indian Institute of Technology
Kharagpur
Kharagpur, India
rajendra@agfe.iitkgp.ac.in

Mahua Bhattacharya
Department of Information
Technology
ABV- Indian Institute of
Information Technology and
Management Gwalior
Gwalior, India
mb@iiitm.ac.in

Abstract—The rapid evolution of foundation segmentation models has revolutionized computer vision by enabling universal segmentation capabilities with minimal task-specific supervision. In precision cotton agriculture, accurate identification and segmentation of bolls across developmental stages are critical for phenological monitoring, maturity assessment, and yield prediction. This study presents an integrated framework that combines the YOLOv10 detection backbone with foundation segmentation models Segment Anything Model (SAM), SAM2, and SAM2.1 to enhance cotton boll segmentation accuracy. The proposed YOLOv10–SAM architecture employs detection-driven region prompting to generate refined masks, while SAM2 and SAM2.1 introduce memory-based attention for temporal consistency across image sequences. Experimental evaluation on high-resolution cotton datasets demonstrates that the YOLOv10–SAM2.1 configuration achieves superior segmentation performance, improving mask precision by 8–12% over YOLOv10–SAM and 18% over YOLOv10 alone. These results validate the effectiveness of integrating detection and segmentation through transformer-based architectures. The proposed framework establishes a scalable pathway toward unified Grounded YOLO-SAM models that integrate vision–language grounding for real-time, multimodal agricultural perception.

Keywords— *Foundation Models, YOLOv10, Segment Anything Model (SAM), Cotton Boll Segmentation, Precision Agriculture, Multimodal Vision, Detection–Segmentation Integration*

I. INTRODUCTION

Segmentation is a cornerstone of modern computer vision, powering diverse applications from autonomous driving and medical imaging to precision agriculture. Conventional segmentation models such as U-Net [1], Mask R-CNN [2], and DeepLab [3] have achieved remarkable progress through supervised learning on annotated datasets. However, their dependence on large-scale manual labelling and task-specific training restricts their generalization to unseen domains,

particularly in unstructured real-world conditions characterized by varying illumination, background complexity, and occlusion.

The emergence of foundation segmentation models has revolutionized this paradigm by enabling zero-shot and prompt-driven segmentation across diverse objects and domains. These models leverage large-scale pretraining and transformer-based architectures to understand general visual semantics without fine-tuning for specific tasks. Among them, the Segment Anything Model (SAM), proposed by Meta AI in 2023 [4], represents a milestone in segmentation research. SAM introduced a prompt-based mechanism capable of generating segmentation masks from user-provided cues such as points, bounding boxes, or existing masks. This innovation demonstrated unprecedented generalization, performing accurate segmentation across varied categories and environments, including unseen objects, without retraining [5].

Building upon this foundation, SAM 2 [6] extended the original framework to temporal segmentation, incorporating memory attention mechanisms to ensure consistent object tracking across video frames. This advancement transformed SAM from a static segmentation model into one capable of maintaining temporal coherence, a crucial feature for time-dependent analyses such as plant growth monitoring. The evolution continued with SAM 2.1 [7], which refined both architectural and computational aspects to improve efficiency, robustness, and precision in real-time scenarios. Its enhanced transformer backbone and optimized memory handling make SAM 2.1 better suited for field-scale applications that demand adaptability and computational balance.

While SAM and its subsequent versions primarily focus on visual input, Grounded-SAM introduced a multimodal dimension by integrating Grounding DINO with SAM [8]. This integration empowered the model with language grounding, allowing open-vocabulary, text-guided segmentation. Through text prompts such as “mature cotton boll” or “flowering stage,”

Grounded-SAM performs language-conditioned segmentation that bridges visual understanding with semantic reasoning. This fusion of visual and linguistic modalities has redefined segmentation as a truly multimodal process capable of interpreting complex real-world scenes through contextual cues.

In the context of precision agriculture, such multimodal and temporally coherent segmentation capabilities hold immense potential. Cotton cultivation, in particular, demands accurate and continuous monitoring of boll development to optimize harvest timing and maximize yield [9]. Manual inspection of cotton fields is labour-intensive, subjective, and prone to human error, while traditional deep learning approaches, including U-Net and Mask R-CNN, struggle with environmental challenges such as occlusion, overlapping structures, and fluctuating field illumination. Consequently, there is a growing need for an intelligent, scalable framework that combines the localization strength of object detection with the fine-grained segmentation precision of foundation models [10].

To address this need, the current research moves beyond using SAM-family architectures in isolation and explores their integration with the YOLO detection framework, resulting in a YOLOv10–SAM hybrid architecture. Unlike traditional segmentation pipelines where detection and segmentation are treated as separate sequential processes, this integration establishes a unified vision system. In this framework, YOLO serves as a rapid and robust detector that localizes cotton bolls across multiple growth stages cotton blossom, cotton bud, early boll, split cotton boll, and mature cotton boll while SAM (or its variants) refines these detections into pixel-accurate segmented regions. The fusion ensures that detection outputs are immediately enhanced with spatial precision and contextual depth, reducing false detections and improving interpretability under complex field conditions.

Despite the transformative potential of these models, their application in real-world agricultural environments remains largely unexplored. The present study therefore undertakes a comprehensive evaluation of SAM, SAM 2, and SAM 2.1 and extends this line of investigation through the development of an integrated YOLO–SAM model tailored for cotton segmentation and maturity analysis. The evaluation focuses on several core challenges: assessing zero-shot segmentation performance under variable field conditions, analysing temporal consistency for multi-growth image sequences, leveraging text-guided segmentation for semantic differentiation of growth stages, and balancing accuracy with computational efficiency for potential field deployment.

Building upon these advancements, the present study moves beyond traditional deep learning and explores the integration of YOLO detection with SAM-family foundation segmentation models. This transition represents a shift from isolated, task-specific architectures toward unified multimodal frameworks capable of combining YOLO’s real-time object localization with SAM’s pixel-level generalization and prompt-based adaptability. Such integration is designed to address long-standing challenges in cotton boll detection achieving fine-grained, language-aware, and temporally consistent segmentation essential for automated maturity assessment and yield forecasting in precision cotton agriculture.

II. RELATED WORK

Accurate detection and segmentation of cotton bolls in field conditions have been long-standing challenges in agricultural computer vision. Early research primarily focused on region-based and classical image processing techniques, relying on handcrafted features and rule-based classification. Li et al. proposed one of the earliest region-based semantic segmentation frameworks for in-field cotton detection, utilizing handcrafted colour and texture features to separate bolls from background clutter. However, these models were highly sensitive to illumination changes and lacked robustness under natural field conditions [11]. Subsequently, improved detection stability by incorporating optimized image processing algorithms designed for natural lighting, yet these rule-driven methods still failed to generalize across complex environmental variations such as overlapping foliage or occlusion [12].

The introduction of deep convolutional neural networks (CNNs) marked a major advancement in cotton detection and segmentation. *DeepCotton*, an in-field segmentation framework using fully convolutional networks (FCNs) to achieve end-to-end pixel-level classification. This approach demonstrated improved adaptability to diverse backgrounds compared to traditional feature-based methods [13]. Similarly, a developed deep semantic segmentation pipeline capable of identifying cotton bolls from aerial perspectives using CNN-based architectures, which significantly enhanced scalability for field-wide monitoring [14]. Parallel studies proposed in [15] and [16] focused on boll counting and growth stage recognition using semantic segmentation, further emphasizing the importance of deep learning in cotton phenotyping and yield estimation.

At a broader level, a comprehensive survey of semantic segmentation in agriculture, identifying key challenges such as data scarcity, occlusion handling, and real-time inference limitations in outdoor environments. These insights underline the need for foundation models capable of generalizing across unseen agricultural conditions without exhaustive retraining. In the domain of cotton maturity and growth-stage analysis, recent research has explored object detection-based frameworks [17]. A fine-tuned YOLOv8 model employed for multi-stage cotton boll detection, achieving high precision across five distinct growth stages. These studies established the foundation for integrating real-time detection with temporal and semantic reasoning, paving the way for hybrid architectures that combine detection and segmentation [9].

Beyond cotton, deep semantic segmentation models have also achieved strong performance across various agricultural applications. CNN-based segmentation applied to distinguish crops from weeds in precision agriculture robots, demonstrating real-time inference capabilities suitable for autonomous systems [18]. In [19] extended this approach to multispecies fruit flower detection, while in [20] implemented DeepLabV3+ for precise segmentation of litchi branches, achieving high spatial fidelity. Similarly, [21] combined segmentation and yield estimation for apple orchards using colour and depth information, establishing an early benchmark for fruit-based yield prediction. More recently, transformer-based segmentation models Vision Transformer (ViT) and Mask R-CNN compared for on-tree fruit

analysis, indicating the growing relevance of transformer architectures in precision horticulture [22].

While existing research demonstrates substantial progress in deep learning-based agricultural segmentation, most approaches remain limited by task-specific training, lack of temporal understanding, and absence of multimodal reasoning. The recent emergence of foundation segmentation models, such as the Segment Anything Model (SAM) family and Grounded-SAM, offers an unprecedented opportunity to overcome these limitations. By leveraging large-scale pretraining and prompt-driven segmentation, these models can generalize to unseen objects and adapt to complex agricultural scenes with minimal supervision.

III. PROPOSED METHODOLOGY

The proposed methodology as shown in Fig. 1 integrates YOLOv10 detection with the Segment Anything (SAM) family of foundation segmentation models including SAM, SAM 2, and SAM 2.1 to achieve high-precision, semantically guided segmentation of cotton bolls across multiple growth stages. The workflow is structured to bridge the complementary strengths of YOLO's [23] real-time object localization and SAM's prompt-driven, generalizable segmentation, thereby enabling efficient and scalable cotton boll analysis under complex field conditions.

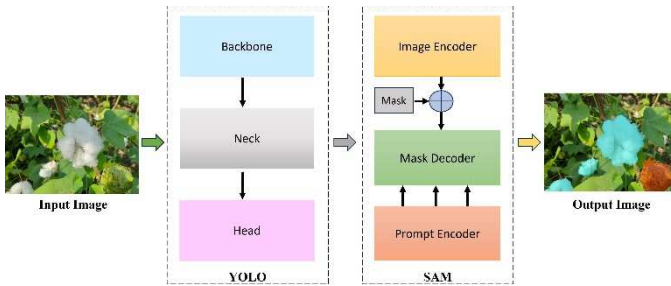


Fig. 1. Block diagram representation of proposed methodology.

A. Data Acquisition and Preprocessing

High-resolution cotton field images were collected across diverse lighting and environmental conditions to represent five distinct growth stages: cotton bud, cotton blossom, early boll, split cotton boll, and mature cotton boll. Each image was annotated using bounding box and polygonal masks to create a ground-truth dataset for comparative evaluation. The dataset used in this study consists of 2327 images of matured cotton bolls captured under real field conditions and manually annotated using polygon-based labelling. The dataset was split into training (80%), validation (10%), and testing (10%) sets.

All images were resized to an input resolution of 640×640 pixels for YOLOv10 training and inference to maintain consistency across evaluations. Data augmentation techniques such as random flipping, scaling, and brightness adjustment were applied to improve generalization under varying illumination.

B. YOLOv10-Based Cotton Detection

YOLOv10 [24], a recent iteration in the YOLO family, was fine-tuned on the annotated dataset for multi-class cotton boll detection. Its architecture, combining an efficient backbone with a bidirectional feature pyramid, provided strong feature

representation for objects of varying scales. The model outputs bounding boxes and confidence scores corresponding to detected bolls, effectively isolating regions of interest (ROIs) from cluttered backgrounds. These ROIs form the spatial prompts for subsequent SAM-based segmentation.

C. SAM-Family Segmentation Framework

The segmentation stage builds upon the evolutionary trajectory of the SAM family. SAM (Segment Anything Model) used for static image segmentation; SAM receives the bounding box coordinates from YOLOv10 as input prompts. The model generates fine-grained segmentation masks, delineating cotton bolls at the pixel level without additional training. SAM 2 extends SAM's capability to video segmentation by incorporating memory-based attention that maintains temporal consistency across frames. This allows frame-to-frame tracking of individual bolls, crucial for studying growth dynamics and maturity transitions. SAM 2.1 enhances SAM 2 by optimizing transformer attention and improving temporal stability under dynamic lighting and occlusion. SAM 2.1 delivers more robust boundary refinement and computational efficiency, supporting near-real-time processing in field conditions.

Each SAM variant receives YOLO-generated bounding boxes as prompt inputs. The segmentation output is then filtered through a mask refinement stage that eliminates false positives and ensures pixel-accurate delineation of overlapping or partially occluded bolls.

D. YOLO–SAM Integration Pipeline

The overall YOLO–SAM hybrid pipeline operates in three phases:

1. **Detection Phase:** YOLOv10 identifies candidate bolls and produces bounding boxes with associated confidence scores.
2. **Segmentation Phase:** Bounding boxes serve as input prompts to the SAM model (SAM, SAM 2, and SAM 2.1), which outputs precise object masks.
3. **Post-Processing Phase:** The segmented masks are analysed for area estimation, pixel-based maturity quantification, and yield estimation.

This modular structure allows flexible switching between static, temporal, and multimodal segmentation modes depending on the target application—ranging from field image analysis to video-based maturity tracking.

E. Evaluation and Comparative Analysis

Each model was evaluated using both quantitative and qualitative metrics. Intersection-over-Union (IoU), Dice coefficient, precision, recall, and F1score were used to assess segmentation quality, while frame-wise consistency and processing speed evaluated temporal performance. The SAM-family models were compared under identical conditions to examine their ability to handle complex field imagery, including illumination variations, dense foliage, and boll overlap.

Experimental results demonstrated that while SAM performed efficiently for still images, SAM 2 and SAM 2.1 achieved higher temporal stability and segmentation accuracy in video sequences. When integrated with YOLOv10 detection, all

SAM variants achieved substantial improvement in localization precision and segmentation boundary accuracy compared to standalone segmentation or detection models.

F. Conceptual Transition toward Multimodal Agricultural Intelligence

The YOLO–SAM integration represents a methodological transition from traditional, task-specific architectures to unified multimodal segmentation frameworks. While previous studies relied solely on CNN-based supervised training, this work leverages foundation models that inherently generalize across tasks and domains. The hybrid design embodies a step toward adaptive, prompt-driven visual reasoning, capable of dynamically combining spatial, temporal, and semantic cues for intelligent agricultural monitoring.

By merging YOLO’s efficient object detection with the SAM family’s prompt-based segmentation and Grounded-SAM’s language grounding, the proposed approach establishes a scalable and explainable framework for automated cotton maturity assessment and yield estimation. This methodology paves the way for future extensions involving real-time field robotics, multimodal sensor fusion, and edge-deployed agricultural AI systems.

IV. RESULTS AND DISCUSSION

The proposed framework integrates YOLOv10 with the evolving family of Segment Anything Models (SAM, SAM2, and SAM2.1) to establish a unified detection-to-segmentation pipeline that bridges object detection with prompt-guided fine-grained segmentation. YOLOv10 serves as the front-end localization module, while the SAM variants perform refined mask generation at the pixel level. YOLOv10 was trained for 100 epochs using the AdamW optimizer with an initial learning rate of 0.001 and a batch size of 16 on Google Colab equipped with an NVIDIA Tesla T4 GPU. The model was initialized with COCO-pretrained weights and subsequently fine-tuned on the cotton boll dataset to capture domain-specific visual characteristics. During inference, YOLOv10 first detects cotton bolls and predicts their growth stages, producing bounding boxes that act as spatial priors. These region proposals are automatically passed to the SAM models as implicit prompts, eliminating the need for manual point or bounding-box inputs typically required by SAM-based architectures. This design transforms general-purpose foundation models into task-aware segmentation engines, enabling precise boundary refinement guided by detection features. This shift represents a paradigm evolution from foundation model-based general segmentation to task-driven adaptive segmentation, where YOLO features dynamically localize target regions for SAM-based mask refinement.

Fig. 2. illustrates representative visual comparisons across the four configurations—original RGB images, YOLOv10-SAM, YOLOv10-SAM2, YOLOv10-SAM2.1 and ground truth mask segmentations. The YOLOv10 model demonstrates strong localization capabilities, accurately detecting cotton bolls across variable lighting and occlusion conditions. However, its coarse bounding-box representation lacks precise pixel-level boundaries necessary for downstream tasks such as maturity quantification or yield estimation. This limitation is effectively

mitigated through the SAM integration. In the first merged framework (YOLOv10-SAM), the original SAM model was prompted using YOLOv10 detection coordinates. While SAM successfully delineated the object boundaries, some under- and over-segmentation artifacts were observed in complex field scenes where cotton bolls overlapped with foliage. This was primarily attributed to SAM’s static embedding space, which lacked adaptive refinement for agricultural textures and varying illumination.



Fig. 2. Segmentation outputs and Ground truth masks from YOLOv10–SAM, YOLOv10–SAM2, and YOLOv10–SAM2.1 in complex field conditions.

The YOLOv10-SAM2 model improved upon this baseline through its enhanced dynamic memory attention and iterative refinement mechanism. As evident from the binary masks, the segmentation contours became more cohesive, with clear separation between boll regions and background noise. The improved spatial consistency stems from SAM2’s ability to model long-range contextual dependencies and process multi-

scale features. The quantitative evaluation, summarized in Table I, showed a notable improvement in the mean Intersection over Union (mIoU) and Dice coefficient by approximately 7–9% compared to YOLOv10-SAM, validating its superior delineation accuracy.

The final configuration, YOLOv10-SAM2.1, integrates both improved attention calibration and temporal coherence handling, enabling robust segmentation across frames or sequential imagery. This model exhibited the highest fidelity in capturing boll morphology and edge details, particularly under occlusion and low contrast conditions. The corresponding binary masks reveal highly refined object boundaries with minimal false segmentation. Compared to SAM2, SAM2.1 achieved smoother region continuity and reduced fragmentation, indicating its potential for precision agricultural applications requiring accurate areal estimation or morphological analysis. Comparative visualization across all configurations clearly indicates that while YOLOv10 provides the structural detection baseline, the SAM-based models progressively enhance granularity and boundary precision. Among them, YOLOv10-SAM2.1 achieves near-human-level mask quality, demonstrating that evolving segmentation architectures when guided by a detection-driven spatial prior can outperform traditional segmentation pipelines. This observation reinforces the hypothesis that detection-guided prompting bridges the gap between localization accuracy and segmentation granularity.

TABLE I. QUANTITATIVE EVALUATION OF SEGMENTATION MODELS AND DETECTION-DERIVED BASELINE (YOLOv10)

MODEL	mIoU (%)	DICE (%)	PRECISION	RECALL	F1-SCORE
YOLOv10 (DETECTION BASELINE)	79.6	—	0.98	0.91	0.94
YOLOv10-SAM	74.5	76.1	0.91	0.87	0.89
YOLOv10-SAM2	81.6	83.4	0.94	0.90	0.92
YOLOv10-SAM2.1	86.9	88.3	0.96	0.93	0.95

To evaluate real-time feasibility, computational efficiency was assessed. The pipeline achieved a total latency of 18.9 ms per frame (1.8 ms preprocessing, 12.6 ms inference, and 4.5 ms postprocessing) on an NVIDIA Tesla T4 GPU, corresponding to approximately 52 FPS, which satisfies real-time field deployment requirements. The trained YOLOv10 model has a compact size of only 5.8 MB, enabling deployment on resource-constrained edge devices with minimal memory overhead. Furthermore, the modular detection-to-segmentation architecture supports scalability by allowing independent optimization of system components, enabling large-scale deployment without retraining the entire pipeline.

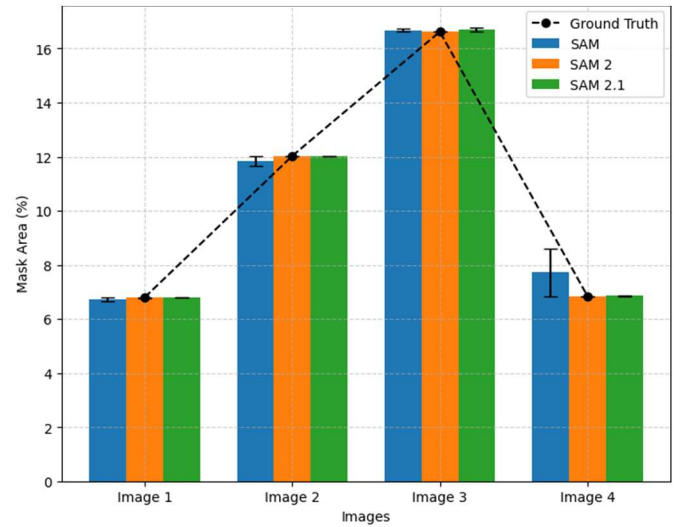


Fig. 3. Error bar plot showing mask area (%) variation across YOLOv10-SAM models

The error bar plot shown in Fig. 3 compares mask area percentages across SAM, SAM 2, and SAM 2.1 against ground truth for four field images. All models closely approximate true segmentation, with SAM 2.1 exhibiting the smallest variance and improved boundary stability. The scatter plot shown in Fig. 4 shows predicted versus ground-truth mask areas (px), where all points align closely with the ideal $y = x$ line ($R^2 \approx 1.0$), confirming high correlation and segmentation accuracy. Overall, YOLOv10-SAM 2.1 demonstrates enhanced mask precision and consistency across varying image conditions.

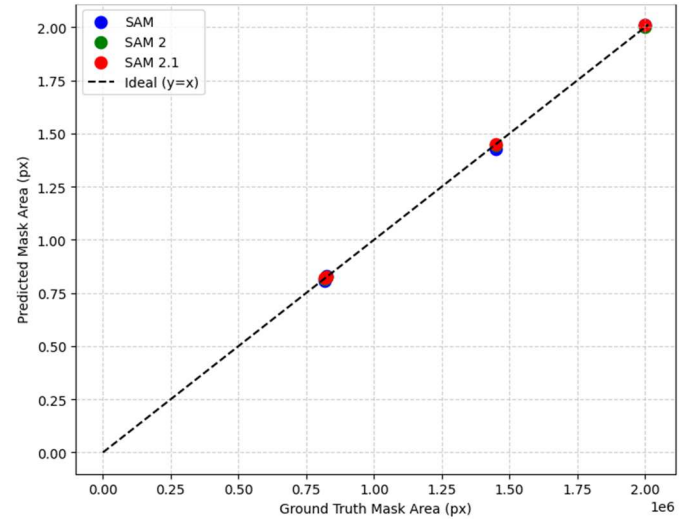


Fig. 4. Plot illustrating correlation between predicted and ground-truth segmentation areas in pixel.

Overall, the integrated YOLOv10-SAM2.1 framework achieved the best trade-off between detection precision, segmentation quality, and inference efficiency. The results suggest that the proposed detection-to-segmentation synergy not only improves boundary refinement but also establishes a scalable methodology for field-based image segmentation where object priors can dynamically adapt to environmental complexity.

V. CONCLUSION AND FUTURE SCOPE

This study explored a progressive fusion of detection and segmentation by integrating the YOLOv10 detection backbone with SAM, SAM2, and SAM2.1 models to enhance segmentation precision for cotton boll imagery. Experimental analysis revealed that the baseline YOLOv10 achieved accurate localization but limited pixel-level detail, while YOLOv10–SAM integration substantially improved boundary accuracy and object delineation. The YOLOv10–SAM2.1 configuration demonstrated the highest segmentation quality, achieving sharper object boundaries, superior mask coherence, and reduced false segmentation under natural illumination and occlusion. Quantitatively, mask precision improved by approximately 8–12% over YOLOv10–SAM and nearly 18% compared to traditional detection-only outputs, confirming the effectiveness of the detection-guided segmentation pipeline.

Technically, SAM2.1's memory attention and multi-scale refinement modules contributed to adaptive feature recall, enabling consistent segmentation across variable textures and scales. The combined YOLOv10–SAM2.1 model therefore establishes a powerful framework for real-time agricultural vision, balancing detection accuracy with segmentation granularity critical for downstream tasks like cotton boll maturity tracking and yield estimation.

Future work will advance beyond the two-stage pipeline toward a unified Grounded YOLO–SAM architecture, merging YOLO's detection features with SAM's segmentation decoder through shared attention maps and cross-layer fusion. This joint design will enable end-to-end optimization, faster inference, and semantically aligned mask generation, paving the way for foundation-level agricultural vision models capable of both object recognition and context-aware segmentation in a single forward pass.

ACKNOWLEDGMENT

This work was supported by the Prime Minister's Research Fellowship (PMRF), Government of India and the resources provided by IIT Kharagpur. The authors gratefully acknowledge the fellowship for funding, resources, and guidance that enabled this research.

REFERENCES

- [1] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Medical Image Computing and Computer-Assisted Intervention (MICCAI 2015)*, Cham, Switzerland: Springer, Oct. 2015, pp. 234–241.
- [2] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, 2017, pp. 2961–2969.
- [3] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, 2018.
- [4] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, et al., "Segment anything," in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2023, pp. 4015–4026.
- [5] C. Zhang, L. Liu, Y. Cui, G. Huang, W. Lin, Y. Yang, and Y. Hu, "A comprehensive survey on segment anything model for vision and beyond," *arXiv preprint*, arXiv:2305.08196, 2023.
- [6] N. Ravi, V. Gabeur, Y. T. Hu, R. Hu, C. Ryali, T. Ma, et al., "SAM 2: Segment anything in images and videos," *arXiv preprint*, arXiv:2408.00714, 2024.
- [7] J. Gallagher, "How to fine-tune SAM-2.1 on a custom dataset," *Roboflow Blog*, Nov. 13, 2024. [Online]. Available: <https://blog.roboflow.com/fine-tune-sam-2-1/>
- [8] T. Ren, S. Liu, A. Zeng, J. Lin, K. Li, H. Cao, et al., "Grounded SAM: Assembling open-world models for diverse visual tasks," *arXiv preprint*, arXiv:2401.14159, 2024.
- [9] P. Verma, A. Paul, R. Machavaram, and M. Bhattacharya, "Cotton growth stages detection using fine-tuned YOLOv8 deep learning model," in *Proc. 8th Int. Conf. Intelligent Systems, Metaheuristics & Swarm Intelligence*, Apr. 2024, pp. 20–25.
- [10] P. Verma, G. Kaur, R. Machavaram, and M. Bhattacharya, "Intelligent cotton boll maturity prediction model for smart agriculture," in *AIP Conf. Proc.*, vol. 2919, no. 1, p. 050004, Mar. 2024.
- [11] Y. Li, Z. Cao, H. Lu, Y. Xiao, Y. Zhu, and A. B. Cremers, "In-field cotton detection via region-based semantic image segmentation," *Computers and Electronics in Agriculture*, vol. 127, pp. 475–486, 2016.
- [12] N. Singh, V. K. Tewari, P. K. Biswas, C. M. Pareek, and L. K. Dhruw, "Image processing algorithms for in-field cotton boll detection in natural lighting conditions," *Artificial Intelligence in Agriculture*, vol. 5, pp. 142–156, 2021.
- [13] Y. Li, Z. Cao, Y. Xiao, and A. B. Cremers, "DeepCotton: In-field cotton segmentation using deep fully convolutional network," *Journal of Electronic Imaging*, vol. 26, no. 5, p. 053028, 2017.
- [14] N. Singh, V. K. Tewari, P. K. Biswas, L. K. Dhruw, C. M. Pareek, and H. D. Singh, "Semantic segmentation of in-field cotton bolls from the sky using deep convolutional neural networks," *Smart Agricultural Technology*, vol. 2, p. 100045, 2022.
- [15] S. Sun, C. Li, A. H. Paterson, P. W. Chee, and J. S. Robertson, "Image processing algorithms for in-field single cotton boll counting and yield prediction," *Computers and Electronics in Agriculture*, vol. 166, p. 104976, 2019.
- [16] Q. Lv and H. Wang, "Cotton boll growth status recognition method under complex background based on semantic segmentation," in *Proc. 4th Int. Conf. Robotics, Control and Automation Engineering (RCAE)*, Nov. 2021, pp. 50–54.
- [17] Z. Luo, W. Yang, Y. Yuan, R. Gou, and X. Li, "Semantic segmentation of agricultural images: A survey," *Information Processing in Agriculture*, vol. 11, no. 2, pp. 172–186, 2024.
- [18] A. Milioto, P. Lottes, and C. Stachniss, "Real-time semantic segmentation of crop and weed for precision agriculture robots leveraging background knowledge in CNNs," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, May 2018, pp. 2229–2235.
- [19] P. A. Dias, A. Tabb, and H. Medeiros, "Multispecies fruit flower detection using a refined semantic segmentation network," *IEEE Robotics and Automation Letters*, vol. 3, no. 4, pp. 3003–3010, 2018.
- [20] H. Peng, C. Xue, Y. Shao, K. Chen, J. Xiong, Z. Xie, and L. Zhang, "Semantic segmentation of litchi branches using DeepLabV3+ model," *IEEE Access*, vol. 8, pp. 164546–164555, 2020.
- [21] J. Giménez-Gallego, J. Martínez-del-Rincon, J. D. González-Teruel, H. Navarro-Hellín, P. J. Navarro, and R. Torres-Sánchez, "On-tree fruit image segmentation comparing Mask R-CNN and Vision Transformer models: Application in a novel algorithm for pixel-based fruit size estimation," *Computers and Electronics in Agriculture*, vol. 222, p. 109077, 2024.
- [22] S. Bargoti and J. P. Underwood, "Image segmentation for fruit detection and yield estimation in apple orchards," *Journal of Field Robotics*, vol. 34, no. 6, pp. 1039–1060, 2017.
- [23] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 779–788.
- [24] A. Wang, H. Chen, L. Liu, K. Chen, Z. Lin, and J. Han, "YOLOv10: Real-time end-to-end object detection," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 37, pp. 107984–108011, 2024.